5.6 Graphing Linear Inequalities in Two Variables

Name _____ Period ____ Date _____

Algebra I 5.6 · Student sheet

I can…

- check solutions of linear inequalities in two variables.
- graph linear inequalities in two variables.
- write and use linear inequalities for real-life problems.

Vocabulary · fill in the missing word

Linear inequality in two variables — _____

Solution — an ordered pair that makes the _____ true

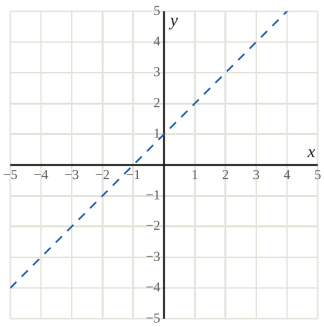
Boundary line — _____ for < or >, solid for ≤ or ≥

Half-plane — the side of the boundary line that contains the _____

Part 1 · Test the plane

Test each point in **y < x + 1**. Plot the points that work in one color and the others in another color.

Point	(0, 0)	(2, 1)	(3, −2)	(−2, −3)	(0, 1)	(1, 4)	(−3, 0)	(−1, 2)	(4, 4)	(2, 3)
Works?										



Pattern: _____

Part 2 · Notes and examples (fill in each note, then work the example)

Example 1

Is (2, −3) a solution of 3x − 2y ≥ 12?

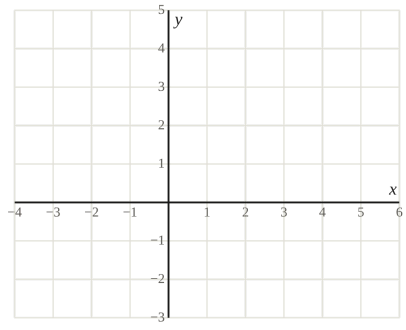
Boundary line: Boundary line: _____ for < or >, _____ for ≤ or ≥.

Test point: Test a point not on the line. If it is true, shade _____ side.

Shading: y > mx + b: shade _____; y < mx + b: shade _____.

Example 2

Graph $y \le -\frac{1}{2}x + 2$.



Example 3

Graph $2x - 3y > 6$.

Example 4

You can spend at most $30 on movies ($10 each) and snacks ($3 each). Write an inequality. Can you buy 2 movies and 4 snacks?

Part 3 · Partner practice

Partner 1

Is $(-1, 4)$ a solution of $y < -3x + 2$?

Partner 2

A dashed line passes through (0, −1) and (1, 1), and the region above it is shaded. Write the inequality.